MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD

ELECTRICAL ENGINEERING DEPARTMENT

Class Test-2 (Session 2025-26) (SET-1)

Course: B. Tech
Subject: Fundamentals of Electrical Engineering
Subject Code: BEE-101

Semester: 1st

Section: E,F,G,H
Time: 60 Minutes
Max. Marks: 20

S. No.	1	2	3	4	5	6
CO No.	CO4	CO4	CO4	CO3	CO5	CO5
Bloom's Level	K1	K3	K3	K3	K2	K3

Note:

1) This paper contains three sections, Section (A) (B) & (C) 2) All sections are compulsory

Section (A)

Attempt this question. This question carries 2 marks. (1*2 = 2)

Q1) What is the role of commutator and brushes in DC machine.

OR

Derive the formula for EMF generated in DC machine.

Section (B)

Attempt all questions. Each question carries 3 marks. (2*3 = 6)

Q2) A 4-pole, long shunt, lap wound generator supplies 25 kW at a terminal voltage of 500 V. The armature resistance is 0.03 ohm, series field resistance is 0.04 ohm and shunt field resistance is 200 ohm. The brush drop may be taken as 1 V. Determine the emf generated.

Q3) Why single phase induction motor is not self starting? Also explain working of shaded pole motor.

OR

Explain the working principle of 3-phase induction motor.

A 4-pole, 3-phase, 50 Hz induction motor has a full-load slip of 4%. Calculate full load speed of motor.

OR

Explain the working principle of synchronous motor

Section (C)

Attempt all questions. Each question carries 4 marks. (3*4 = 12)

Q4) In a 25 kVA, 2000 V/200 V transformer the iron and copper losses are 200 W and 400 W respectively. Calculate the efficiency at half-full load and 0.8 power factor lagging. Determine also the maximum efficiency and the corresponding load.

OR

Define the term voltage regulation in transformer. Determine voltage regulation of transformer whose ohmic drop is 2.25 % and leakage reactance drop is 7. 5 % at 0.8 of lagging.

Q5) (a) What is the need of earthing in electrical installation? Write the names of different methods of earthing.

(b) Write 2 advantages & 2disadvantages of XLPE cable over PVC cables.

OR

Complete equations of Lead Acid & Nickel Iron batteries.

Q6) (a) A battery has taken a charging current of 5.2. A for 24 hours at a voltage of 1.8 V. while discharging it gave a current of 4.5 A for 24 hours at an average voltage of 1.48V. Calculate the ampere-hour efficiency of the battery.

(b) Write a short note on MCB.

For any query contact to: Mr Saurabh Saxena 9456031300/7895043195, Mr. Tanuj Kr. Bisht 9149273074

approved

20.11.25

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MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD

ELECTRICAL ENGINEERING DEPARTMENT

Class Test-2 (Session 2024-25) (SET-1)

Course: B. Tech | Semester: 2nd | Section: A,B,C,D

Subject: Fundamentals of Electrical Engineering | Time: 60 Minutes

Subject Code: BEE-201 | Max. Marks: 20

S. No.	1	2	3	4	5	6
CO No.	CO3	CO5	CO4	CO3	CO4	CO5
Bloom's Level	K2	K2	K2	K3	K3	K3

Note:

1) This paper contains three sections, Section (A) (B) & (C) 2)All sections are compulsory

Section (A)

Attempt this question. This question carries 2 marks. (1*2 = 2)

Q1) What are the different types of losses in transformer?

Section (B)

Attempt all questions. Each question carries 3 marks. (2*3 = 6)

Q2) Write a short notes on ELCB & MCB.

OR

Write advantages of PVC cable and XLPE cable over other cables.

Q3) Draw torque slip characteristics of 3 phase induction motor. A 3 phase, 4 pole 50 Hz induction motor has a full-load slip of 4%. Calculate full-load speed of motor.

OR

Explain the working principle of Synchronous motor. Also write applications of synchronous motor.

Section (C)

Attempt all questions. Each question carries 4 marks. (3*4 = 12)

Q4) (a) The maximum efficiency of a 100 kVA transformer is 98.40% and operates at 90% full load unity power factor. Calculate the efficiency of a transformer at unity power factor at full load.

OR

Derive the formula for emf equation of normal 2 winding Transformer. Also write working principle of transformer.

Q5) (a) A 20 kW, 200 V shunt generator has an armature resistance of 0.05 ohm and a shunt field resistance of 200 ohm. Calculate the power developed in the armature when it delivers rated output.

OR

Derive EMF equation of DC machine.

i) protect appliance

(b) Derive the formula for torque developed in DC machine.

OR

ii) safety

Explain double revolving field theory. Write the names of different starting methods of 1-phase I.M.

Saftey

Q6) What is the need of earthing in electrical installation? Write short notes on pipe earthing.

OR

Explain working of Lead Acid & Nickel Iron batteries by writing Complete equations mentioning active materials.

Pipe Earthing

Rod Earthing

For any query contact to: Mr Saurabh Saxena 9456031300/7895043195

approved

Saurabh Sax
19.06.25

magnetic field Rotar
constant speed.
Efficiency.